

Calculations in Drug Adherence

Gowtham A. Rao

2025-08-31

Contents

1	Introduction	1
2	Key Concepts	1
3	The Calculation Methodology: Dose Era Construction	2
3.1	Step 1: Data Extraction and Preparation	2
3.2	Step 2: Identifying Gaps and Overlaps	2
3.3	Step 3: Constructing Dose Eras	2
3.4	Step 4: Calculating Adherence (PDC/CMA)	2
4	Comparison with Medication Possession Ratio (MPR)	2
5	Illustrative Example	3
6	Conclusion	3

1 Introduction

Medication adherence measures the extent to which patients follow their prescribed treatment regimens. The OHDSI **DrugAdherence** package provides a standardized and efficient framework for calculating adherence using data in the OMOP Common Data Model (CDM), primarily utilizing the **DRUG_EXPOSURE** table.

This vignette describes the methodology implemented in the package, focusing on the construction of Dose Eras to calculate the Proportion of Days Covered (PDC) or Continuous Medication Availability (CMA).

2 Key Concepts

Understanding the following concepts is essential for using the **DrugAdherence** package:

- **Observation Window:** The specific time period during which adherence is measured (e.g., 365 days following the first prescription/index date).
- **Days Supply:** The duration a prescription is expected to last, sourced from the **DRUG_EXPOSURE.DAYS_SUPPLY** field.
- **Gaps and Overlaps:** Gaps are periods between the end of one supply and the start of the next. Overlaps (stockpiling) occur when a patient refills a prescription early.
- **Permissible Gap (Grace Period):** A crucial parameter defining the maximum number of days allowed between the end of one prescription and the start of the next without considering the treatment discontinued. This is often configurable (e.g., **CMA_gap_days**).
- **Dose Eras:** Continuous periods of medication availability constructed by merging individual drug exposure records based on the permissible gap.

3 The Calculation Methodology: Dose Era Construction

The core calculation in the `DrugAdherence` package relies on constructing Dose Eras. This approach calculates the Proportion of Days Covered (PDC), which is robust to stockpiling and inherently capped at 100%. The process is implemented using efficient SQL queries.

The calculation involves the following steps:

3.1 Step 1: Data Extraction and Preparation

The package extracts relevant records from the `DRUG_EXPOSURE` table for the target patient cohort and the specified medications (ingredients or drug concepts) within the defined Observation Window. Records are sorted chronologically by `DRUG_EXPOSURE_START_DATE`.

3.2 Step 2: Identifying Gaps and Overlaps

The SQL logic uses window functions (specifically the `LAG` function) to compare the end date of one exposure with the start date of the subsequent exposure for the same patient and medication.

```
-- Conceptual SQL snippet illustrating gap identification
LAG(DRUG_EXPOSURE_END_DATE) OVER (PARTITION BY PERSON_ID, DRUG_CONCEPT_ID ORDER BY DRUG_EXPOSURE_START_DATE) AS PREVIOUS_END_DATE,
LAG(DRUG_EXPOSURE_START_DATE) OVER (PARTITION BY PERSON_ID, DRUG_CONCEPT_ID ORDER BY DRUG_EXPOSURE_START_DATE) AS GAP_DAYS
```

- If `GAP_DAYS` is positive, there is a gap in therapy.
- If `GAP_DAYS` is zero or negative, the exposures are contiguous or overlapping (an early refill).

3.3 Step 3: Constructing Dose Eras

Dose Eras are constructed by merging consecutive exposures based on the defined Permissible Gap (e.g., `CMA_gap_days`).

1. **Handling Overlaps:** If `GAP_DAYS` ≤ 0 , the exposures overlap. The subsequent exposure is merged into the current era. The era end date is the maximum end date of the merged exposures. This ensures that overlapping days are not double-counted.
2. **Handling Gaps:** If $0 < \text{GAP_DAYS} \leq \text{CMA_gap_days}$, the gap is considered acceptable. The gap period is assumed to be covered (e.g., the patient used remaining supply or the gap is clinically insignificant), and the subsequent exposure is merged into the current era.
3. **Breaking Eras:** If `GAP_DAYS` $> \text{CMA_gap_days}$, the gap is too large. The current Dose Era ends, and the subsequent exposure marks the start of a new Dose Era.

This process results in a set of non-overlapping, continuous periods where the patient is considered to be covered by the medication.

3.4 Step 4: Calculating Adherence (PDC/CMA)

Once the Dose Eras are constructed, the adherence calculation is performed:

$$PDC = \frac{\text{Sum of the duration of all Dose Eras within the Observation Window}}{\text{Total Days in Observation Window}}$$

The result is a value between 0 and 1 (or 0% and 100%).

4 Comparison with Medication Possession Ratio (MPR)

It is important to distinguish the PDC calculated via Dose Eras from the simpler Medication Possession Ratio (MPR).

- **MPR Formula:** (Sum of Days Supply for all fills) / (Total Days in Observation Window)
- **MPR Limitation:** MPR simply sums the DAYS_SUPPLY. If a patient stockpiles medication by refilling early, the MPR can exceed 100%, which is often misleading.

The Dose Era construction method used in the **DrugAdherence** package avoids this limitation by focusing on the actual time covered rather than the total supply dispensed.

5 Illustrative Example

Consider a 90-day Observation Window, with a Permissible Gap set to 5 days.

Fill	Start Date	Days Supply	End Date	Gap from Previous	Action
1	Day 1	30	Day 30	N/A	Start Era 1
2	Day 25	30	Day 54	-5 (Overlap)	Merge into Era 1
3	Day 60	30	Day 89	6	Gap > 5 days. End Era 1. Start Era 2.

Calculations:

1. **Era 1:** Starts Day 1. Fill 2 overlaps Fill 1. In standard OHDSI era construction, the merged era runs from the earliest start date (Day 1) to the latest end date (Day 54) of the merged exposures. *Era 1 Duration:* 54 days.
2. **Era 2:** Starts Day 60, Ends Day 89. *Era 2 Duration:* 30 days.
 - **Total Days Covered:** $54 + 30 = 84$ days.
 - **PDC:** $84 / 90 = 0.933$ (93.3%).
 - **MPR:** $(30 + 30 + 30) / 90 = 1.0$ (100%).

(Note: The exact handling of the end date when gaps occur can vary slightly based on implementation details, but the core concept of merging based on the permissible gap remains the same.)

6 Conclusion

The OHDSI **DrugAdherence** package implements a robust, SQL-based methodology for calculating medication adherence (PDC/CMA) through the construction of Dose Eras. This approach aligns with modern adherence standards by correctly handling stockpiling (capping adherence at 100%) and allowing for clinically acceptable gaps in therapy, providing a standardized measure across the OHDSI network.